

PROJECT NAME: Automotive Diagnostic And Activity Monitoring System (ADAM)

TEAM NAME: PROJECT ADAM

Abstract

ADAM system for automotive diagnostic and activity monitoring is an innovative vehicle monitoring apparatus that ensures road safety while response takes place in case of vehicle failure and also serves an investigation purpose. ADAM serves different purposes such as monitoring of critical parameters that include RPM, coolant temperature, throttle position, and diagnostic trouble codes (DTCs). It stores data locally in memory and transfers data securely via the internet to a remote database.

Based on technologies of Raspberry Pi, machine learning, and mobile app integration, this state-of-the-art diagnostic solution provides real-time analysis of driving behavior, generates driving reports, and presents opportunities for the accumulation of data. It provides the owner with timely information to enable proactive maintenance, optimum vehicle performance, and prevention of the emergence of a bigger problem.

It's worthy of mention that ADAM works in forensic analysis, through collecting data and analyzing it for investigations post-accident. The system provides law enforcement, insurance companies, and vehicle owners with critical details to enhance transparency and minimize fraud in the automotive repair sector.

Thus, with its twin purpose of preventive maintenance and forensic analysis, the ADAM System represents an evolutionary step in the advancement of modern automotive management solutions that positively influence safer roads and more efficient vehicle operation.

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